Optima Academy Online

An Education Experience Company

Calculus Honors

Course Syllabus · Fully Asynchronous · Full Year

COURSE DESCRIPTION

Calculus is the mathematics of change and of the infinitely small. For two thousand years geometry could measure straight lines and static shapes but stumbled on the moving and the curved. Calculus is the instrument humanity finally built to reason about change itself: the exact speed of a falling body at a single instant, the slope of a curve at one point, and the area beneath a curve that no simple formula describes. In this Honors course you will master the two great instruments of the subject, the derivative and the integral, and see how a single idea, the limit, makes both possible. The work is demanding, precise, and genuinely beautiful. You are joining a conversation that runs from Archimedes through Newton and Leibniz to the mathematics that runs the modern world.

WHAT YOU WILL LEARN (YEAR AT A GLANCE)

Quarter	Focus
Quarter 1	Limits and continuity; the definition and meaning of the derivative.
Quarter 2	Rules of differentiation, including the chain rule, implicit and inverse functions, higher-order derivatives, and the Mean Value Theorem.
Quarter 3	Applications of the derivative (curve analysis, optimization, related rates, motion) and the integral (Riemann sums and the Fundamental Theorem of Calculus).
Quarter 4	Applications of the integral (area, average value, volume) and an introduction to differential equations.

HOW THE COURSE WORKS

This course is fully asynchronous and delivered through Canvas. The year is organized into four quarters, each with four modules that run two weeks apiece.

- **Each week** has two lessons to work through, one problem set you complete by hand and upload, and one short auto-graded quiz.
- **Each module** also includes a discussion, one Module Challenge for your fullest and most careful work, and one VR activity.

- **Each module opens** with a short intro survey and **closes** with an outro survey. Complete the intro first and the outro last; they help your teacher see where you are.

MATERIALS AND TECHNOLOGY

- A dedicated math notebook and pencil. Much of your learning happens in your own handwritten work.
- A graphing calculator or a free graphing tool, for checking estimates against a picture.
- Reliable access to Canvas and to a Microsoft 365 account for any typed submissions.
- A VR headset for the module VR activities (desktop and mobile access are available with reduced features).

GRADING

Your grade reflects three kinds of work, weighted to reward steady practice, thoughtful participation, and careful demonstrations of understanding.

Category	Weight	What it includes
Practice and Skill (auto-graded)	50%	Weekly quizzes and weekly problem sets (scored on completion and good-faith effort). VR activities are included here on completion.
Discussion and Reflection	20%	Module discussions and the intro/outro surveys.
Module Challenges	30%	One substantive, teacher-graded task per module, graded on a consistent four-part rubric.

> These weights are a recommended default; your section may adjust them slightly. The Module Challenge rubric is published separately and stays the same all year, so you always know what excellent work looks like.

ACADEMIC INTEGRITY AND THE USE OF TOOLS

Graphing tools and computer algebra systems can compute a limit or a derivative in an instant, and you are welcome to check your work with them. They cannot do the part that matters most in this course: understanding why a method works and choosing the right one. That judgment is what you are building, and it cannot be outsourced. Submit your own reasoning, always, and show your work so that your thinking is visible.

HOW TO SUCCEED

Calculus rewards steady, careful work far more than last-minute effort. Keep a neat notebook, work every practice problem before checking it, and write your solutions vertically with your reasoning

shown. When you get stuck, note exactly where, and bring that question to office hours. A little each day beats a great deal all at once.